PharmaLeaf · B.Pharm Semester VII · PCI 2026 (NEP 2020)

BP701T Biostatistics and Research Methodology

PharmaLeaf notes for B.Pharm Semester VII (PCI 2026, NEP 2020). Written in easy language, unit by unit in syllabus order. Every formula is shown in a blue box and every topic has a **worked pharmaceutical example** (all numbers checked by calculation). Symbols are written in words where the handwriting font has no Greek letter: **mu** = population mean, **sigma** = population standard deviation, **x-bar** = sample mean, **s** = sample standard deviation, **n** = sample size.

Syllabus at a glance

BP701T is a 3-credit core theory course: 3 lectures per week, 45 hours in total, 75 marks (30 sessional + 45 end-semester).

Unit	Title	Topics	Hours
I	Basic concepts of biostatistics	Variables; data collection and cleaning; population and sample; probability and non-probability sampling; descriptive and inferential statistics; frequency distribution; central tendency; dispersion; degrees of freedom; quartiles, skewness, kurtosis; diagrams	9
II	Probability and probability distributions	Classical and statistical probability; union, intersection, complement, conditional and marginal probability; PMF and PDF; binomial, Poisson, normal and standard normal distributions; sampling distributions: t, F, chi-square	9
III	Correlation and regression	Pearson and Spearman correlation; scatter plots; multiple correlation; simple linear regression (least squares), coefficient of determination; multiple regression; overfitting and underfitting	9
IV	Inferential statistics	Point and interval estimates; confidence intervals; hypothesis testing; type I and II errors; power; p values; t tests; one-way and two-way ANOVA; tests in regression; Mann-Whitney, Wilcoxon, Kruskal-Wallis, Friedman, chi-square tests	12

Unit	Title	Topics	Hours
V	Research methodology	Research types and designs; research methodology steps; decision tree for selecting tests; descriptive, observational and experimental designs; report writing, plagiarism, referencing, journals and databases; screening and optimisation designs (Plackett-Burman, factorial, D-optimal, simplex, CCD, RSM), blocking and confounding	6

Unit I: Basic Concepts of Biostatistics (9 hours)

I.1 Definition and meaning

Statistics: the science of **collecting, organising, summarising, analysing and interpreting numerical data** to draw conclusions and make decisions under uncertainty.

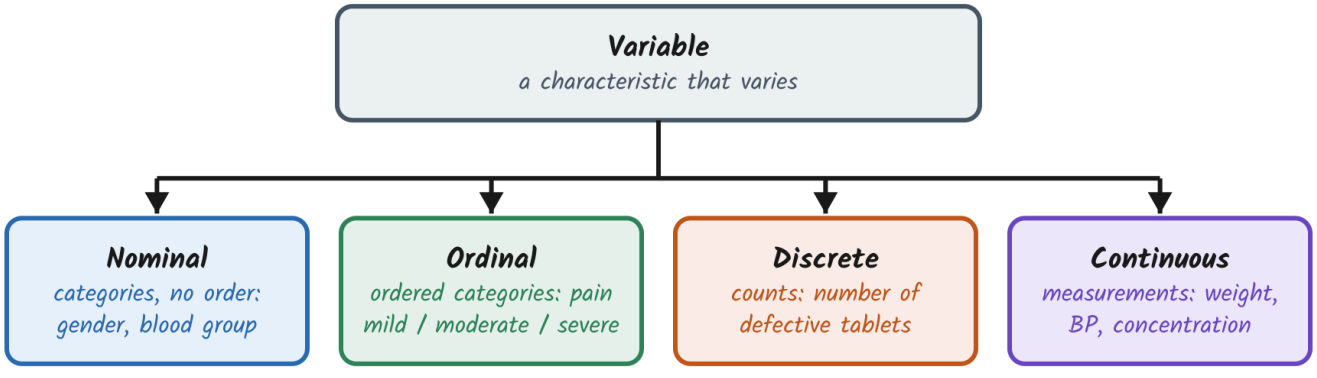
Biostatistics: application of statistical methods to **biological, medical, pharmaceutical and public-health** problems: clinical trials, bioequivalence, quality control of batches, stability studies, epidemiology, pharmacovigilance, laboratory experiments.

Uses in pharmacy: design and analysis of **clinical trials**; **BA/BE** studies (90% confidence intervals); **quality control** (weight variation, content uniformity, control charts); **stability** and shelf-life (regression); **formulation optimisation** (factorial designs); **pharmacology** (dose-response, ED50); **epidemiology** (prevalence, risk); **pharmacovigilance** (signal detection).

I.2 Variables and their types

Variable: a characteristic that **varies** from one person or object to another (age, blood pressure, tablet hardness, gender).

Types of Variables



Remember: nominal + ordinal = qualitative; discrete + continuous = quantitative

Scales: nominal, ordinal, interval (no true zero), ratio (true zero)

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Type	Meaning	Examples
Qualitative (categorical)	Describes a **quality or category**	
Nominal	Categories with **no natural order**	**Gender**, blood group, drug class, route of administration
Ordinal	Categories with a **meaningful order**, but unequal gaps	**Pain score (mild, moderate, severe)**, cancer stage, ADR severity
Quantitative (numerical)	Measured or counted **numbers**	
Discrete	**Countable whole numbers**	**Number of tablets defective**, number of ADRs, number of patients
Continuous	Can take **any value** in a range (measured)	**Weight, height, blood pressure, plasma concentration, dissolution %**

Scales of measurement: nominal, ordinal, interval (equal intervals, **no true zero**: temperature in °C), **ratio** (equal intervals and a **true zero**: weight, concentration, dose).

By role in a study: independent variable (cause, controlled or manipulated: dose, formulation), **dependent variable** (outcome measured: blood pressure fall, % release), **confounding variable** (related to both: age, smoking), **control variables** (kept constant).

1.3 Data: meaning and methods of collection

Data: a collection of **facts, observations or measurements** recorded for analysis. **Primary data:** collected **first-hand** by the investigator for the study. **Secondary data:** already collected by others (hospital records, published literature, government surveys such as **NFHS**, PvPI databases).

Method	Description	Example
Direct observation / measurement	Investigator measures directly	Tablet weights, BP readings
Experiments	Controlled conditions	Animal studies, dissolution
Interviews	Face-to-face or telephone with a schedule	Patient adherence
Questionnaires / surveys	Structured forms (paper or online; Likert scales)	Knowledge-attitude-practice (KAP) surveys of pharmacists
Records and registries	Existing documents	Case records, prescriptions, ADR forms